Install and configure PyTorch on your machine.

Article • 07/01/2024

ⓘ **Note**

For greater functionality, [PyTorch can also be used with DirectML on Windows](#).

In the previous stage of this tutorial, we discussed the basics of PyTorch and the prerequisites of using it to create a machine learning model. Here, we'll install it on your machine.

Get PyTorch

First, you'll need to setup a Python environment.

We recommend setting up a virtual Python environment inside Windows, using Anaconda as a package manager. The rest of this setup assumes you use an Anaconda environment.

1. Download and install Anaconda here . Select `Anaconda 64-bit installer for Windows Python 3.8`.

ⓘ **Important**

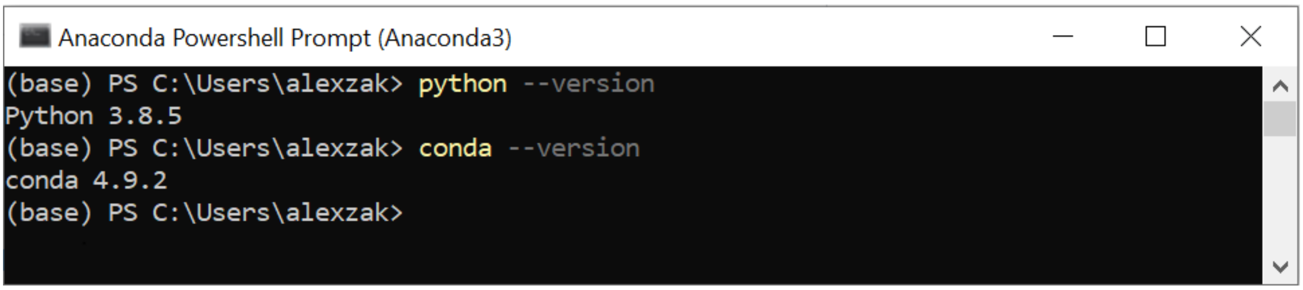
Be aware to install Python 3.x. Currently, PyTorch on Windows only supports Python 3.x; Python 2.x is not supported.

After the installation is complete, verify your Anaconda and Python versions.

2. Open Anaconda manager via Start - Anaconda3 - Anaconda PowerShell Prompt and test your versions:

You can check your Python version by running the following command: `python --version`

You can check your Anaconda version by running the following command: `conda --version`

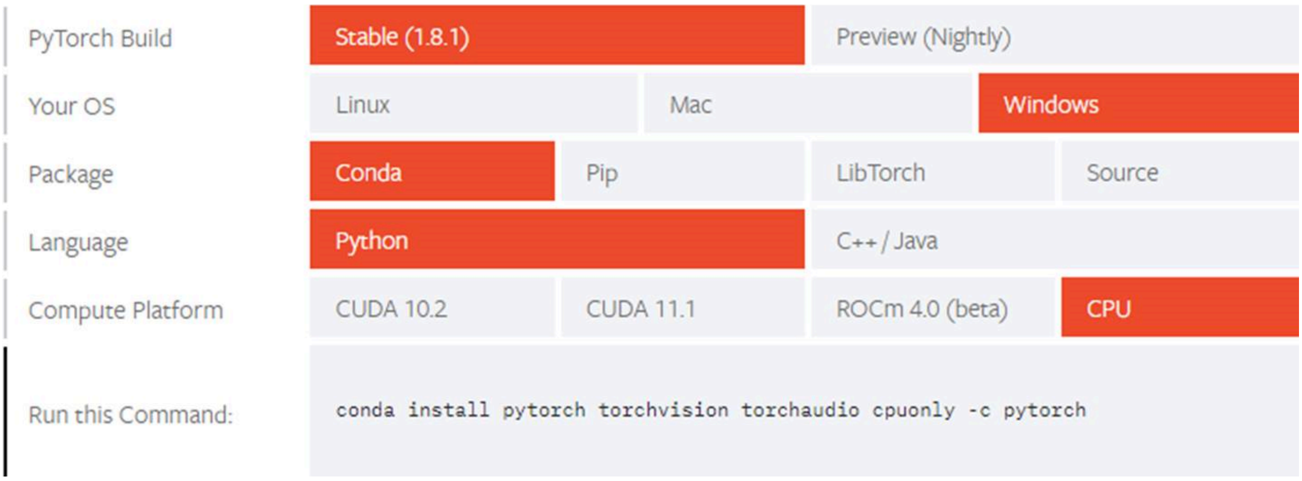


Now, you can install PyTorch package from binaries via Conda.

3. Navigate to https://pytorch.org/ .

Select the relevant PyTorch installation details:

- PyTorch build – stable.
- Your OS – Windows
- Package – Conda
- Language – Python
- Compute Platform – CPU, or choose your version of Cuda. In this tutorial, you will train and inference model on CPU, but you could use a Nvidia GPU as well.



4. Open Anaconda manager and run the command as it specified in the installation instructions.

```
conda install pytorch torchvision torchaudio cpuonly -c pytorch
```

Anaconda Powershell Prompt (Anaconda3)

```
(base) PS C:\Users\alexzak> conda install pytorch torchvision torchaudio cpuonly -c pytorch
Collecting package metadata (current_repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\alexzak\Anaconda3

  added / updated specs:
    - cpuonly
    - pytorch
    - torchaudio
    - torchvision


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    conda-4.9.2                |   py37haa95532_0         2.9 MB
    cpuonly-1.0                |                0           2 KB  pytorch
    libuv-1.40.0               |       he774522_0         255 KB
    ninja-1.10.2               |   py37h6d14046_0         246 KB
    pytorch-1.7.1              |      py3.7_cpu_0       157.0 MB  pytorch
    torchaudio-0.7.2           |             py37         2.7 MB  pytorch
    torchvision-0.8.2          |          py37_cpu        6.6 MB  pytorch
    typing_extensions-3.7.4.3  |             py_0          28 KB
    ------------------------------------------------------------
                                           Total:       169.7 MB

The following NEW packages will be INSTALLED:

  cpuonly            pytorch/noarch::cpuonly-1.0-0
  libuv              pkgs/main/win-64::libuv-1.40.0-he774522_0
  ninja              pkgs/main/win-64::ninja-1.10.2-py37h6d14046_0
  pytorch            pytorch/win-64::pytorch-1.7.1-py3.7_cpu_0
  torchaudio         pytorch/win-64::torchaudio-0.7.2-py37
  torchvision        pytorch/win-64::torchvision-0.8.2-py37_cpu
  typing_extensions  pkgs/main/noarch::typing_extensions-3.7.4.3-py_0

The following packages will be UPDATED:

  conda                                4.8.2-py37_0 --> 4.9.2-py37haa95532_0


Proceed ([y]/n)?
```

5. Confirm and complete the extraction of the required packages.

Anaconda Powershell Prompt (Anaconda3)

```
Proceed ([y]/n)? y


Downloading and Extracting Packages
pytorch-1.7.1        | 1007.0 MB | ################################################################
########### | 100%
torchaudio-0.7.2     | 2.7 MB    | ################################################################
########### | 100%
libuv-1.40.0         | 255 KB    | ################################################################
########### | 100%
torchvision-0.8.2    | 7.3 MB    | ################################################################
########### | 100%
ninja-1.10.2         | 247 KB    | ################################################################
########### | 100%
cudatoolkit-11.0.221 | 627.0 MB  | ################################################################
########### | 100%
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
(base) PS C:\Users\alexzak>
```

Let's verify PyTorch installation by running sample PyTorch code to construct a randomly initialized tensor.

6. Open the Anaconda PowerShell Prompt and run the following command.

```
python
```

Next, enter the following code:

```
import torch

x = torch.rand(2, 3)

print(x)
```

The output should be a random 2x3 tensor. The numbers will be different, but it should look similar to the below.

Anaconda Powershell Prompt (Anaconda3)

```
(base) PS C:\Users\alexzak> python
Python 3.8.5 (default, Sep  3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> x = torch.rand(2,3)
>>> print(x)
tensor([[0.5343, 0.7362, 0.8390],
        [0.4758, 0.2309, 0.1936]])
>>>
```

> **Note**
>
> Interested in learning more? Visit the PyTorch official website

Next Steps

Now that we've installed PyTorch, we're ready to set up the data for our model.

Feedback

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